

Product Flow Map v4

The complete reference for the Deal Origination Platform · every rule, flow and safeguard as built · 22 July 2026

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System Map & Data Stores

One BigQuery dataset is the single source of truth. Every page is a view of the same tables; nothing is ever copied between pages.

THE STORES (DATASET: AVERROES_DEAL_FLOW)

TABLE	HOLDS	WRITTEN BY
targets	Every company: identity, financials (source-tagged), CH intelligence, scores, outreach state, action buckets, IC memos	Uploads, SmartFill/Enrich, email sync, stage moves
investors	LP universe: funds, angels, family offices, agencies — typed, scored, staged	Portfolio mining, PitchBook LP upload, InvestorFill
investor_links	Connection layer: one row per investor↔company relationship (type, stake, evidence)	Investor miner (daily)
email_log	Every email exchanged with known contacts, full history, deduped by Message-ID	Email sync (regular + deep)
activity_log	Timestamped audit trail per company: stage moves, notes, enrichment summaries, migration markers	Everything
qualification_config	Versioned qualification criteria	Criteria editor

THE FOUR SURFACES

Deal Pipeline

kanban of targets by stage

Master Universe

the full table, filters, bulk ops

Investors (LPs)

LP table, mining, outreach

Intelligence Chat

data-grounded Q&A over everything

Doctrine: UI pages are views, never owners of state · no second copy of any fact · event timestamps record when things happened, not when we processed them · description only ever replaced by longer text.

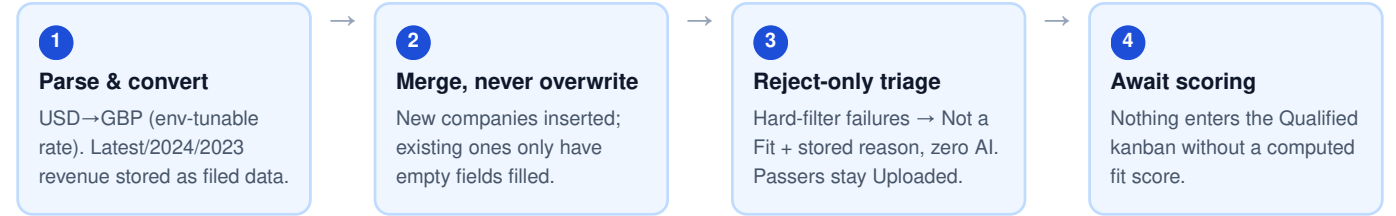
Sourcing & Ingestion

Companies enter the universe from scrapers, PitchBook exports and Inven exports. Ingestion costs zero AI — intelligence is spent only on companies worth it.

CHANNELS

CHANNEL	WHAT ARRIVES	STATUS ON ARRIVAL
Scrapers (EF alumni, Tech Nation, SaaStock...)	Name, site, sector basics	Scraped
PitchBook company export	40+ fields: financials, investors, growth predictions	Uploaded
Inven CSV (auto-detected by filename)	Currency-converted revenue (3yr), growth metrics, EBITDA margin, directors, LinkedIn, investors & owners	Uploaded

INVEN PIPELINE



Rule: pre-qualification never promotes. It only rejects clear failures (geography, industry, revenue envelope) so SmartFill's budget is never wasted on them — promotion to Qualified requires a scored SmartFill pass.

Qualification & Fit Scoring

Calibrated to the mandate: £15–40M equity cheques for majority or significant-minority (25%+) stakes in UK/Ireland B2B software.

HARD FILTERS (RUN FIRST, ALWAYS)

1. Geography — UK or Ireland

2. Industry — B2B software / tech-enabled

3. Revenue envelope — £2.5–40M investable

Failures are gated to Not a Fit with a hover reason before any grounded AI runs — the single biggest cost guard.

REVENUE BANDS V3

Too Early < £2.5M

Target Band £2.5–40M

Too Large > £40M

Scoring curve peaks in the £8–20M sweet spot (4–6x EV/ revenue implies the cheque buys 25–100% there).

FIT SCORE: FIVE WEIGHTED METRICS

Revenue size	band-curve position
Revenue growth	local CAGR first, web judgement fallback
Employee growth	local Inven data OVERRIDES web guesses
Business fit	thesis alignment (grounded)
Market sentiment	press/awards/hiring evidence (grounded)

REVENUE ESTIMATION (WHEN UNFILED)

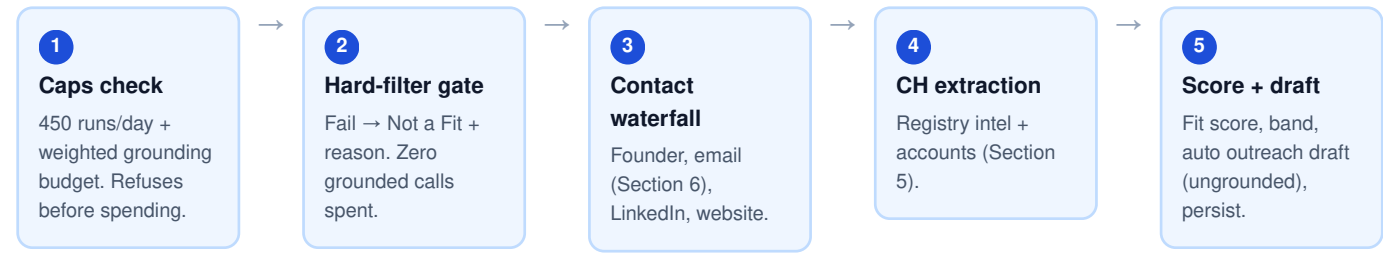
Median of proxies: employees × £100K, assets × sector multiple, gross-profit override, cash rule — blended with a grounded Gemini estimate, stored with confidence + source. Estimates never masquerade as filed data.

Deal math anchor: valuation ≈ 4–6x revenue; at the 5x midpoint the £15–40M cheque buys 25%+ anywhere in the target band. This arithmetic is computed in code wherever it appears (scoring, IC memos) — never narrated by AI.

SmartFill Anatomy & Data Precedence

The expensive first pass: qualify, enrich, extract, score, draft — in strict cost order.

EXECUTION ORDER



DATA PRECEDENCE (ENFORCED IN THE PERSIST LAYER)

FIELD FAMILY	RULE
Financial figures	CH filed value wins; NULL from a failed parse never wipes stored Inven/PitchBook figures. Dates follow their number.
Contacts / website / LinkedIn	Non-blank finding wins; blank keeps stored.
CH registry fields	Written only when CH matched this run; failed lookups wipe nothing.
Description	Longer wins.
Scores / band	Always recomputed fresh.

BULK MODE

100 companies/run through `POST /smartfill/batch` : the server processes the list in ~2.5-minute windows, streaming a heartbeat every 10s (hostile networks kill silent connections), returning per-company truth + remainder. A 10-minute idempotency guard makes re-sent batches free. Real failures surface in the modal with reasons.

SMARTENRICH (THE CHEAP SIBLING, ~¼ COST)

Always source-verifies the stored contact (confirm / replace / clear) · refreshes registry intel · re-parses accounts only if a newer filing exists (or one-time history backfill) · re-scores if fresh financials · regenerates unsent drafts (sent emails never touched) · writes a timestamped action summary to the activity feed.

SECTION 5

Companies House Intelligence

Three API layers — Public Data, Document, and a scheduled watch — turned into underwriting-grade data.

REGISTRY INTELLIGENCE

Ownership proof: PSC register → verified ownership label + founder %

Distress flags: overdue accounts, insolvency history, registered charges (debt radar)

Filing intelligence: last resolution, accounts regime trajectory — crossing micro→small→medium thresholds is a growth signal

Accounts due: next filing date — timing intel for outreach

ACCOUNTS EXTRACTION (DOCUMENT API)

Multi-year history: up to 6 filed periods — revenue, gross profit, profit, total/net assets, cash, employees — parsed from the actual PDFs (stored in GCS), feeding the profile's charts and multi-year P&L.

Cap table (CS01): holders aggregated across share classes, percentages computed in code on one basis (always sums to 100), Others row, international corporate suffixes recognised for founder detection.

DAILY WATCH

07:00 UTC scheduled sweep of the 80 least-recently-watched pipeline companies; new-filing alerts land in the activity feed; one batched stamp keeps it ~1 minute. The whole pipeline rotates every few days. The LP miner refresh rides the same run.

MATCHING

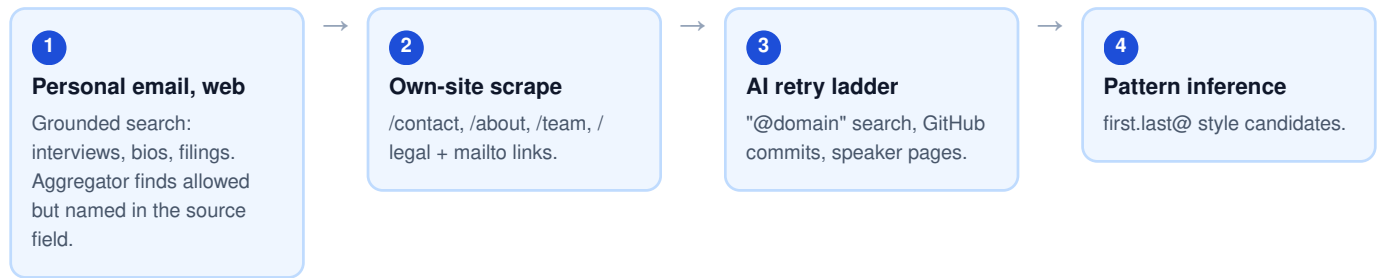
Name search with an exact-core gate and confidence score; low-confidence matches are stored as notes, never as facts.

Why it matters: filed accounts are the most verifiable data source in the system — they win the precedence rules, anchor the deal math, and the CS01 cap tables answer "can we buy 25%+" before the first call.

SECTION 6

Contact Waterfall

Four escalating tactics; a guess is never stored unless the mailbox verifiably exists; blank beats bounce.



Verification gate: mailbox verification (Hunter) runs ONLY for step-4 guesswork and must return strictly deliverable. Found-in-source addresses are accepted as found. A verified personal guess beats a generic fallback (hello@); if nothing survives, the field stays blank.

LIFECYCLE

Runs inside every SmartFill; SmartEnrich re-verifies the stored address on every pass (confirm / replace / clear an unsourceable guess); the whole waterfall was retro-applied to previously enriched companies via marker-guarded startup migrations. The email's provenance is stored beside it.

Outreach Engine

Draft-first, human-send-always. The founder email follows a fixed structure (v7) that reads like a person, not a mail-merge.

EMAIL STRUCTURE V7

1. Hi {first name} + short pleasantry
2. **Who I am** first (Beatrice, Averroes Capital)
3. Follow + conviction, exactly 2 sentences: "We have been following X and really like <specific>. We believe <problem> is a real pain point..."
4. Portfolio proof: "Journey and Glowday"
5. Humility line
6. Ask: "...quick 20-minute call..."
7. "Look forward to hearing back from you." + "Best," alone

RULES & SAFEGUARDS

- Natural company name (registry casing and Ltd stripped)
- Zero em dashes anywhere (the model mimics instruction style)
- Stored-signal hook: reuses scoring evidence as the news hook at zero cost; fresh lookup only within budget
- Draft auto-generated at qualification, refreshed by SmartEnrich, NEVER regenerated over a sent email
- Generation failures retry twice; the emergency fallback template is flagged and never persisted
- Send persists the actual recipient — replies match even if the contact later changes

TEST LOOP

The Internal Test company (source = Internal Test) always sends to admin@, auto-resets to Qualified on removal, and bypasses stage guards — the whole loop is rehearsable end-to-end without touching a real founder.

SECTION 8

Email Sync & Action Buckets

Every sync turns the mailbox into structured intelligence: full history captured, replies classified, and each founder response filed into an action bucket with a drafted answer.

SYNC MECHANICS

Matching: contact email → drafted/sent recipient → company domain (free-mail excluded). **Deep mode:** one IMAP search per corresponded contact over 10 years — complete history, no mailbox-size cap. **Dedup:** Message-ID. **Stage truth:** any reply advances Engaged → Responded; reply timestamps use the email's own Date header; a self-heal pass fixes anything missed.

THE TEN ACTION BUCKETS (OUR DATA × THEIR WORDS)

Right fit — set up call

Right fit — answer & advance

Redirect — follow the referral

Right fit — structure mismatch

Right fit — too early, nurture

Not now — timing

Right fit — too large, stay close

Needs human read

Declined — close politely

Not the right fit — do not respond

One ungrounded call per reply reads the email against the record (band, fit, mandate); founder-stated numbers win. Output: bucket + one-sentence rationale + follow-up date + suggested reply for buckets that warrant one.

Boundaries: buckets never move stages — they recommend, the human decides. Suggested replies are review-and-send only, and a hygiene pass clears them automatically once we are the latest message in the thread. Failures are reported loudly in the sync message.

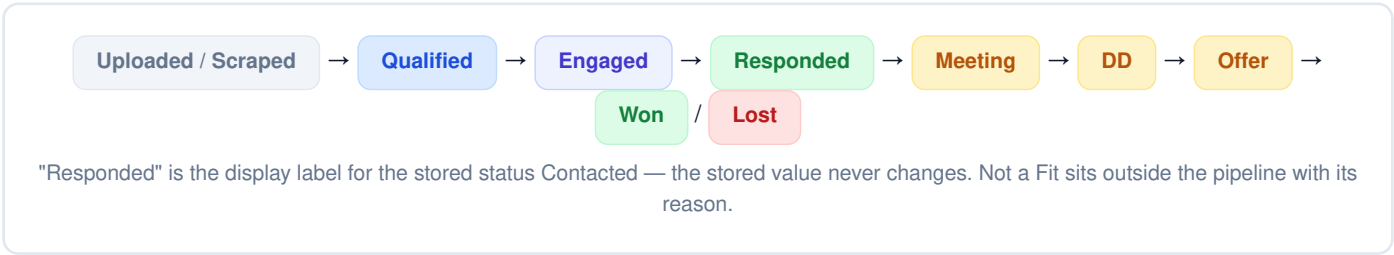
WHERE IT LANDS

The bucket replaces "Email Sent" as the primary button on Responded cards (colour = urgency, hover = rationale, click = Outreach tab); the Responded column auto-sorts act-now first; the profile shows the full panel with the suggested reply and copy button.

Pipeline Lifecycle

Stored stages with display labels; every transition stamped twice — current-stage entry and permanent first-entry.

THE FLOW



STAGE RULES

Enter Qualified	Scored SmartFill pass, or manual "Qualify anyway" on the score
Enter Engaged	Outreach actually sent (draft alone never moves a card)
Enter Responded	Any reply, automatically on sync
Meeting onwards	Human moves (button or drag); stage changes always via the shared endpoint
Timestamps	stage_entered_at resets per move (drives sorting + the 10-day stale flag); per-stage first-entry columns are stamped once, never overwritten
Sorting	Qualified by fit score; Responded by action-bucket priority then recency; others by recency

THE CARD

Identity + reply/stale badges · band + fit badges · financial snapshot · contact · primary action (outreach 3-state, or the action bucket once responded) · IC memo button from Engaged onwards · advance/note/remove. Clicking the name opens the full-screen profile (7 tabs, arrow-key browsing).

IC Memo

A committee-ready one-pager per deal, generated in ~20 seconds. Numbers assembled in code from the verified record; AI writes only the prose; gaps say "Not yet known".

THE NINE SECTIONS

1 Opportunity	What they do, sourced proprietarily, why now
2 Mandate fit	Three checks with PASS / FAIL / UNKNOWN + evidence from the record
3 Financial snapshot	Source-tagged table: CH filed / Inven / estimate + confidence
4 Deal hypothesis	4–6x valuation range + what the £15–40M cheque buys (computed)
5 Ownership & cap table	Founder %, CS01 holders, PSC verification
6 Engagement	From the actual email thread
7 Market context	ONE grounded search, sources named inline
8 Risks & open questions	Registry red flags injected automatically + what the first meeting must answer
9 Recommendation	One sentence

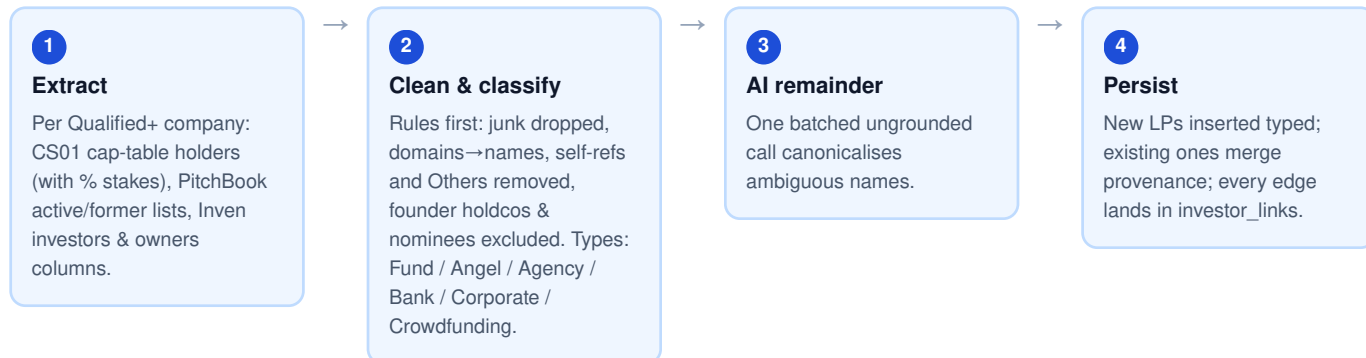
MECHANICS

Button on Engaged+ kanban cards and the profile's IC Memo tab · stored as JSON on the company row (regenerate re-reads the latest record) · Download PDF renders an A4 one-pager for the IC pack · cost: one grounded call (weight 1).

LP Module & Connection Layer

A second database of potential LPs — and a graph of who backs whom across the universe.

PORTFOLIO MINING V2 (DAILY, AUTOMATIC)



THE CONNECTION LAYER

`investor_links` : one row per investor↔company relationship (link type, stake, evidence). Queries it answers: all investors of a company · all portfolio companies of an investor · co-investors · sibling companies sharing a backer. Surfaced in the profile's Ownership tab ("Investor connections" + "Connected companies") and via `/connections/*` endpoints.

THE REST OF THE MODULE

Sources: mining + PitchBook LP export + public directory scrapers. **InvestorFill:** grounded research per LP (AUM via net-assets proxy, PSC principals, ticket fit) with a 4-metric LP fit score. **Stages:** Identified → Researched → Contacted → Meeting → Committed / Passed, with AI-drafted LP outreach and CSV shortlist export.

Warm-intro rank: an investor's overlap count (how many of our targets they back) is the priority signal — each overlap is both an intro path into a deal and a credibility bridge for the fund raise.

Intelligence Chat

Ask anything about the universe and the LP book in plain language — answers come from the database, never from guesses.

BEHAVIOUR CONTRACT

Data-only answers	Grounded in the full universe index (companies + LPs); if the database does not know, it says so — and never invents
Typo-tolerant matching	"sasscada" finds SaaScaDa — spaceless and distinctive-word matching
Web search on request	When data is missing, the chat offers a user-triggered grounded search (weight 1) rather than silently searching
Consultant format	Short bolded sections and tight bullets, rendered properly in the chat UI
SmartFill nudge	Thin records prompt "run SmartFill on X" instead of speculation

Cost Governance, Ops & Security

Intelligence spend is budgeted like capital: caps first, weights per operation, nothing outside the accounting.

DAILY BUDGETS

SmartFill/Enrich runs	450 / day
Grounded-call budget	1,400 weighted / day
Weights: SmartFill 4 · SmartEnrich 3 · InvestorFill 1 · news lookup 1 · IC memo 1. Every grounded call passes the shared budget check and is logged — no exceptions.	

ZERO-AI BY DESIGN

Ingestion, pre-qualification triage, investor extraction rules, all UI reads, connection queries, deal math, financial tables. Ungrounded (cheap) AI: reply classification, action buckets, chat answers, draft generation, investor-name canonicalisation.

SCHEDULED & OPS ENDPOINTS (TOKEN-GATED)

/ch-watch/run	daily 07:00 UTC — registry watch + LP miner refresh
/enrich-oneoff/run	batched SmartEnrich sweeps
/prequalify/run	reject-only triage sweep
/email/deep-sync/run	full-history mail backfill
/investor-mine/run	LP mining on demand

SECURITY & AUDIT

Google Sign-In with domain allowlist + named exceptions · 12-hour HMAC session tokens · visible sign-out; dead sessions redirect instead of rendering empty pages · secrets live in env vars only (never the repo — enforced after a real leak) · every mutation lands in the activity log with event-true timestamps · migrations are marker-guarded to run exactly once.

Verification doctrine before every push: full compile, real import with route asserts, TypeScript check, cache-busted live checks after deploy. Failures surface loudly in the UI — silent failure is treated as a bug in itself.